Active InferAnt Stream 001.2 ~ FieldSHIFTing for the SoftComp Intelligent Soft Matter Workshop

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so hello everyone uh and welcome to this session of intelligent soft matter so what we try to organize is a workshop uh on very unusual workshop and this and the structure of this Workshop is also very topic is unusual and the structure is unusual uh so this is the website what you see on this screen and uh this is about intelligence of matter and it is not uh it is emerging field so it is not yet well developed and it will be maybe in future one of the direction of the soft matter and since not much it was done in this field what we are planning in this field and what we see what can be done from different perspectives so we have people working with uh materials and soft matter and experimental right and we have people from sensors and nanop particles and how to to um to sense the environment and we have also people from decision making artificial intelligence uh which also a different field so there is a mixture of different profiles and uh today what we wanted to do because the topic is very unusual and new uh here where is um uh a description and uh and also we have a program and all and we have also profiles of the people so today what we wanted to do is uh is to in this meeting should be uh also an unusual format it will have an unusual formats that it is not only presentation of the of the talks but also uh collaboration and initiation of the collaboration and discussion and uh uh we can and to to initiate the this discussion and to make it meaningful uh what we did is uh to we we wanted to analyze the contributions and also that people who will join that they may uh know each other uh before the meeting and also to have some ideas of other people and also of other directions and also other ideas of people what they can uh bring to this meeting and with whom they can meet and what they already if we start writing a paper for example we have in the plan uh the paper on though in those sessions we have a a plan to write a perspective paper and when we write this perspective paper uh we should already prepare I and to to be prepared and to prepare some some uh part of this text and uh to be ready to collaborate with people because still we have very short time during the meeting that's why uh I suggest so we have this session and additionally we have a session on grant writing session and this is uh uh Pro will be um organized by I leag uh and also with u uh this will be a session traditional grant writing and addition and and last day we will have idea generation session which is more open and it will be uh uh prepared which is what we are doing now we will will be prepared in collaboration with uh two guests one is uh Danielle uh fredman from California and he is a director of the active influence Institute and today we have another guest is the Tobias K from Netherlands and he is um uh a s of the component knowledge pixels who is behind of Nano Publications so why this is uh important so we have uh uh we have the uh in here what we we want to do is to generate some ideas before for the meeting this is the the the goal and one side we will use artificial intelligence and uh the modern tools which allowed not only to present your directions but also to get insights from uh other fields through this automatic process and this is this idea generation process through field shift and this will be presented by Daniel fredman and on other side if we have those ideas we can uh register those ideas as small pieces of information with a name and the and this can be a nanop publication which can be in a uh machine readable format and that can also allowed to build on top of this piece of knowledge so the then what we to see how we can merge all those new technologies and to see how it it is possible so first of all I would like maybe to uh that maybe our guests will tell a little bit about themsel and also about this tools so first of all we start with Tobias and then with Daniel will leave maybe like this so Tobias do you want to to present your Tool uh sure yeah so um well I can just very quickly uh introduce myself I'm Tobias Kon I'm actually no longer based in the Netherlands I'm still affiliated with University there but I'm now in Zurich Switzerland and have now we have now our small startup called knowledge pixels and um yeah as um Vladimir already said we are um focusing on this concept the technology of nanop Publications um yeah I know do you want me to now already give a give maybe one or two minute that's a concept of this nanop publication and then we go further because then we will discuss how to include it and after this how to merge it later but if you just uh tell what is the concept how it works and that's will be enough okay yes of course so as the name says an publication is like a publication but tiny um and on top of that it's a publication it's not just written in English or or any natural language but it's uh um written in you could say a logic language if you're a bit familiar with semantic web or link data Technologies it's written in the rdf language which means uh it's made of sentences where you have subject predicate object and all the three of them are globally unique identifiers um so people around the globe can say things and when they talk about the same thing it will automatically connect uh um yeah if I may kind of um simplify it a bit um so nanop

Publications are kind of these tiny Snippets of such statements in the simplest case it's just a single such sentence with something is related with that thing to that thing like I know in the in medicine it could be this this Gene has this this kind of effect on this disease often it's more than one such triple as they are called because often statements are a bit more complex uh than just a simple triple and that together so these couple of statements then consists the main content of nanop publication uh which is called the assertion but then it always has some Provence to it meaning you can state with another such triple where this came from for example because you had a study that delivered it or because it was extracted from an existing paper or data set or whatever and then it has some metadata where you can say who created it with which tool uh with a Tim stamp with a license and these things uh and all that together then is an anop publication it gets a stable persistent identifier of its own and the the thing that we are now working with the thing that we providing is that you're building a decentralized ecosystem system for dealing with such nanop Publications so you can push them to the world basically to the nanop publication Network and it can then live at different places so when you later uh you or somebody else want to retrieve it again you can and because has this formal structure you can also uh ask very specific queries that can then precisely be answered uh for humans to consume but also for machines and for applications to build up on um so we can also on the flight create new classes and ation so we can create new vocabulary and then talk about it and make n Publications you have a tool for that that's called Nano Dash that is supposed to be user friendly so it's still kind of Cutting Edge so we there's still a lot of work ahead of us um but anyway so it's kind of a low threshold to publish these and then you and others can query it again and um yeah we're now in the first steps to kind of apply this more broadly um um across different domains and with connections to journal and yeah there's lot I could talk about more but I think I've already used up my three minutes yeah so that's exactly the purpose of this uh uh tool how to use this tool so it is machine readable format and this is very short and easy to publish and this goes in the direction of the the new new trends of this uh language models then they come and the the process of Publications uh the dissertation right writing projects writing and those public Large review writing that becomes Obsolete and already doesn't have much sense to to have all this uh uh all all this traditional Legacy publishing system so instead of this it it goes in the direction of very uh specific and personalized uh tools and also so in the artificial intelligence now becomes more and more in the it goes in the more and more in the direction of uh agent based so instead of General tools for everything so they have you will have a a agent which will be uh doing some uh work or task very personalized for you and for for other people so you will communicate through the agent and this is how we see how this Publications can enter into this domain and this is how we connect with Daniel from active influence Institute because the whole Institute is focusing on the development of those artificial intelligent tools for that will be uh making those agents which will be making interaction between humans and machine and this is one of the examples how we can apply these two uh uh intelligence uh soft matter to this workshop and we can just show how at this again this is a very early stage but how this can be approached maybe now Daniel you can come up and show this tool and explain any about works great I'm going to share my screen okay okay all right well thank you Vladimir for inviting me and it's been really fun to collaborate on these uh directions also Tobias that's very exciting about the Nano pubs I think broadly uh this is going to be the unstructured text getting into where the structured information of Nano pubs is going to supercharge it but e even today I think there's some pretty fun and exciting at least provocative directions to explore um it's I'm going to push this all to the GitHub link that's in the video description and in the side chat um okay so to kind of back up and then we'll look at this on the open source GitHub last week we had an applied active inference Symposium at The Institute and then in uh those sessions we kind of analogously had participant information presenter abstracts and so on so working with Vladimir we had some discussions and we built some tools around like what can we do to add value and context before the event how can we start to think about templates and transferable tools and practices across different related or overlapping or even totally separate open science and participation initiatives so we made this repo called Symposium so Vladimir made a bunch of uh edits and and commits to the this uh before last week and recently including adding in some open source language model methods what I'm going to show is just a few more researcher side process accelerations perhaps using perplexity that's available through the browser at perplexity Ai and here is using the the API so what I did here is I took the perplexity folder that I had applied to the in active inference Symposium participants and then adapted this in this folder called ISM stream for intelligence soft matter so that's where everything that we're going to talk about today is in here so this is it just as the folder named it just meant different pieces for people to pull out um and explore and then also I'll note that this is all done in cursor especially as there was all already a folder of this working for a different structure but analogous tasks getting to this ISM stream content was was like 10 to 20 total prompts and cursor updates um okay so in the inputs folder Vladimir provided the information on the participants and also there's a cat so here's intelligence soft matter and then there's speakers and participants and they each have this open Alex and this works so the works are some citations and bibliography and the topics are are topics of the person okay so the what the first uh script does is it researches the presenter so everything goes into the outputs folder here's the presenter uh folder so in ad Jason and a markdown here are internet-based researchs all experimental research tooling so it's it's hard to even estimate sometimes it's totally accurate if it's little information in a common name this isn't going to be very useful if there's a ton of information and and links provided it's super useful so this this outputs a summary on the presenters um but I I kind of wanted to explore the uh working with the participant information more so that's what most of the following scripts are so the second script two and all these are run with just P from the ism stream folder just Python 3 and then the name of the script so for the participants started off with also writing uh research report I don't know if we can look up someone who's here now to see see how accurate it is but yeah yeah g is here and it looks quite accurate okay so again sometimes it's super accurate sometimes double check okay so wrote this wrote This research report then this is where field shift comes into play so these are just sort of playing around with different combinatorics of of what is possible and also this is just using all by all folder iterating so it's not using a database it's it's hardly agentic in the sense that we could call any part of a computer pipeline agentic like is a database search a search agent I guess if we do think about it that way then it is but this is just using all by all iterating on these folders which keeps it plain text and and really available also for nanop pubs and GitHub and all this so what field shift does there's a folder called domain so in this folder called domain this is copying a folder out from a previous um repo and live stream called field shift that was based upon some work by Mike L slap so what domain has is it has these synthetic knowledge files so they're just specific questions examples factual statements within a domain so just some of them are they're just meant to load up the context and just provide information on a synthetic domain but you could imagine that could be like a person's record or an organization's record or a different kind of document but here just being used just to say these are just statements and open questions and problems and this happened this year all those kinds of things for different domains so what three field shift does is it takes each each participant's background and it it um will find I didn't run this for every single participant but but we'll just run in cursor now so this takes each participant's background and then at basically uses perplexity to J toose researcher background and the synthetic domain and then asks what could they learn from the field and how could they contribute to the field so this is Daniel craft and RX and fur and then the way that again there's more sophisticated ways to do prompt um engineering but what's Happening Here is it just sending to perplexity here's the participant analyze the bidirectional knowledge exchange between name and participant here's the profile of the participant here's the knowledge domain just copy paste um it each of these take like so here let's just look at this one that just was made here's what this person could learn from RX iner these are accurate parts of of what they could learn from that field how could they contribute to the domain I don't know but that that's from into so that's that's field shift and then that just iterates over whichever domains are in the domain folder so it's like here's statements about William Blake um that's uh three so that's the the um field shift okay then Vladimir provided a folder of catechisms so catechism can mean different things but in the project management setting it's a set of question and answers that are aligned with project management being a grant officer program manager these kinds of questions that are important to be addressing so took this for shifted catechisms here this was like one prompt from three copied three and said okay now with the catechism in at catechism folder write a that structure for each of these shifted fields so now with the Daniel craft blockchain bir directional relationship now it's going to write a catechism with that structure so it'll it'll take a few seconds um then just while this one's processing um the other way to use a c ISM here is for all pairwise participants write a catechism that would reflect a complimentary project using the karma Gap I don't know what it is but that catechism format so we'll we'll look at okay here's five so here's Daniel craft X dong and then that is just saying okay now for these two participants there's other ways that you could fill up the context window and everything but just saying here's how you write the document here's here's what here's what we're looking for with the document Styles here's the template of the catechism here's the participant information here's their other proposals that we know from them here's their own here's their field shifted um to and from and then make a collaborative project okay first let's look at the just the field shift um blockchain I I don't know this person but we'll see you know they they can see to and from blockchain and then just last piece is the collaborative proposal we'll look at it in in GitHub yeah maybe I can also show uh the results when we we were Shifting the profiles of the people because they are coming from different domains uh to the uh context of the workshop it's intelligence soft matter and what was the outcome uh so the were it was done uh with three steps first step is to generate profile of the each person and also the questions uh maybe if I I switch to online now let I I will I will show the outcome of this uh result yes Danielle yes uh okay uh uh this is here so this is also in in the same uh uh giup GitHub folder so if you go here so there is the inputs and inputs are basically they are taken from uh this is intelligence oft matter and participants speakers and we have outcome is the the publication taken from open Alex so there is a publication uh directions research directions uh topics and uh also citations so it's it this is uh provided in in this format so we have uh this as a basis so we don't use any other information and we can do it for any researcher and after this uh the question was uh after we we have threee u s generate profile of the researcher with respect to the field of intelligence of matter then we have a researcher methods we it can extract the methods and also description of the methods from each profile and additionally we can do collaborative projects and the projects depending on the catechism what Daniel explained so what for example we have European projects they have a a certain structure if we have a a American project or some other funding then it will be completely different questions and different structure and it will be a differ so the catechism is here so this just a text file and for example we if we have European grants this is a certain structure if we have and here we we have just another one this is katein for karma projects and this is just the questions what need to be addressed and this is a structure first is a vision and scope project statement proposal solution collaboration knowledge integration plan yes so we use this to construct projects uh with and also connections between people so then and how it works so we have it is in this folder synthetic and here this is the also generated so first of all we have all merged research profiles of all participants in this file so this is a um a automatically generated file and the same we have automatically generated description of the intelligence soft matter and the and this is current understanding future Direction and changes so this is done in this format and this is what is known in general so this is basically uh generated contracts but H you can also have a look so this is also may be interesting to see if you agree or not and this is basically general knowledge available in the internet about this this field and future Direction so this is again uh it is a rapidly evolving with several excited future directions and those are seven directions uh what basically we it is discussed in the literature so it is not coming from anywhere and uh uh uh and we have also open questions this is for fundamental science for computational science uh material design applications ethicals and emerging questions so there are the questions and future Innovation so this is again uh how to to because if we talk in the workshop about intelligence of matter we need to talk what do what do we consider it and this is our actual State maybe after the meeting it will be completely Rewritten we have a new ideas and they will be doing this and here is the beauty of this method so we can be based everything is based on agents and then we can re regenerate all the structure again this is not a problem because uh it takes maybe 10 minutes after this we go to participants and then again we can pick up the participant uh there is a uh first of all so we have a a research profile so the research profile that you already saw uh there is the research focus of the person then we impact analysis so this High cations focus on fundamental corness we have research Evolution from where it started and where this researcher goes and then there is a ke contributions and also future directions with respect to this field intelligence soft matter so uh where this profile can fit in in in this uh structure what we proposed and this and if we will if in ENT oft matter will be redefined this is fine then this all those profiles will be again regenerated and then uh another one is a uh the collaboration so we know this field we know where this researcher come from and then the question is uh what collaboration what is the matching contribution he may search and here what we see that there are several suggestions to whom he may collaborate so it will be complementary and uh so that this is a totally uh automatic uh and there you can also have a look and then and after this uh it a third uh part so we have an second one is uh the pro we have methods so we can ex extract methods from also from Publications and then uh analyzing those publication we can see that there is a list of methods with a description with Monto simulations and then with the description and uh the result outcomes then we have also several models so this is we see all those uh theoretical models and uh we see also emerging Trends how this field evolves and limitations for example if you use scaling models scaling Theory rization group what will will be the limitations and uh this is very useful when we talk about collaboration and projects after this uh we go to Project generation and for example this is the project generated for this particular researcher and here what we see is the emerging computation and con biohybrid materials networks so here we see all description so then then uh starting from the title uh they have project rational and objectives so this we have according to this catm uh what we set up for example if we want to structure it in the in the form of European project there is a special uh setup for European project if it is in this was done for karma Gap and here we see uh the um uh the structure for this particular project and here what we see all those topics like alignment with you you calls project Vision objectives and contribution to PO priorities expected kpis then they have St of of the art gaps and limitations novelty um and then we we see also alignment with Europeans this is uh by the way this is a European structure method methodology what can be done in this methodology expected impact and you added value and the long-term risk management and this all this resources and work plan and deliverables so this is already uh we have this dissemination exploitation so ethics and all these things so this is a European structure so we we have another project uh which was done uh here so this is another project is a which which is called a different type of project with a completely different structure and this is simplified this is karma Gap projects are simplified and here we have a vision and scope we don't have any Europe Dimension gender balance and all this things which will be uh very particularity of the European project here it is a vision and scope why this project needed now a societal need problem statement and uh proposed solution and K Innovation and what is also important that here we also see uh with this project which is another type of project with neuromorph soft met by inspired mive Network ofed nanop particles it just suggests the person from the workshop who will be the best contact with respect to this uh uh project for example here what is this idea so we have a neuromorphic integration so this is what is this so this is M resistors and how to make them from soft matter materials using photocatalytic nanop particles embedded in polymer Matrix so this is the topic and uh uh it was generated so we have a connection with other participant and uh we have a structure already already for the grand proposal and it was generated in 100 seconds so it's basically two 2 minutes and you have a project like this then with a full details and here also what is also included uh Network and collaboration so who else from the meeting can contribute to this idea and here because we know all the profiles so they just list those people from the from the workshop uh and how these people can contribute to this particular project and uh again we have this uh uh methods so this is again what methods need to be taken into account they have this St Sam spectroscopy so computational data biological data machine learning so then it will be uh also the structure of the implementation plan so this is structure of the project we it can be a several year project and how to structure it and uh we have a methodology what what else is necessary and what is missing and you see a the it says also that not all people from the workshop is enough that we will be again uh some extra people this uh material scientist engineer it's missing and we don't have it in the workshop but if we just limit people from the workshop it just give you this uh those names which are more suitable for this project so this is uh the the the way to operate so this is this will be just an example of course we can delete this all and it may not uh maybe not uh it will be just an example so this we can use it as an example of future collaboration so for us it is important that if you can just log in and have a look of researcher profiles it is also possible to generate for uh even for each person to whom to talk and to to which subject for example you can with your profile then you can talk to another person and also present something that maybe you don't even realize that you can work together on something on this and that's maybe we can have a list of people and to topics that you can address those people because they are experts in this field so this is more or less what we are doing now and this is how the the this uh field is developing so and I I know that Daniel want to show something else thank you Vladimir that's awesome that's cool okay I'm going to show the last piece of the script and I mean I hope this is giving some fun ways in each symposia that you all or anyone organizes or or we organize we can keep on iterating and making it more interesting okay let me just share take a second [Music] do you see okay let me try screen sharing again I'll just show this for um a minute than than any any thoughts or questions on this okay H says I can't share my screen I'll try one more time Roger Sherin okay so meanwhile if you have any questions you can ask let me just reload I'll be back in one second interaction and and this is why we we also see a big uh um very big potential in nanop Publications because uh this is the way to structure the pieces of information which is uh relevant and which you can say this because if you build a project what you everything can be generated but what is important is what is really uh verifiable uh data and what is a statement which is a is a you can trust and this is where this nanop publication come because you don't need the whole paper you don't need to write to read reviews but what you need is to to have those pieces that you can construct your own project you can construct your own paper or something like this and this is uh can be also done automatically yes and this is where we see this interaction between AI humans and the this system of the blockchain which is basically uh what the to shows okay um it's it's uh it's a funny thing but just navigate over to the is M stream and I'll show I'll show it on the stream but just for those in the in the chat um in the ism stream there's a stream there's a folder six called visualize um and what what visualize does again it'll be visible on the uh stream I'll run it in Koda 2 okay first thing it does is it iterates over each kind of the output document so the researcher profiles I I know you don't see it in the Google chat but but it's visible on the on the stream but just it just a funny thing is happening but then um it iterates over each kind of document like the research profiles and the other and then it uses standard set of linguistic analysis methods like like topic modeling and principal component analysis all these will be in the in the um visualizations folder and and so it makes for example principal component plots and and what those principal components um are loaded on for different researchers and looks at different topic modeling and statistical aspects of of the semantics of the event so that opens up a lot of really interesting possibilities for like designing unlikely connections and setting up projects that are in certain regions and especially combined with the methods that Vladimir just mentioned Nano pubs I mean having a nano CV with available methods this is really changing the game because of the semantics being openly accessible and the rate limiting step to a large extent being like what projects are really going to happen with who and how uh yes I we see uh the ail first M MH oh okay thanks thanks a lot for this is I got the the general pictures very clear and what I don't have clear is so how can we refine the the results because all these results were obtained by automatic information that you know we're in the web but uh how do you what is the how do you iterate in such a way that you know this can be refined because this could be all junk right in PR yeah yeah yeah of course yes and this is the way that this is just to give an idea of before the meeting for of other participants and then to see what can be done and then uh to uh refine it is basically on the the field what you want to to achieve then uh the structure of of the questions what you need to to answer and uh also the profile method profile so this the basic information then it because what it does is just the shifting so example if you have a method in in polymer physics so and you want to apply it in some maybe some elements of quantum quantum theory into polymer physics so and you know that this is how Deen got to Nobel Prize and then and and he just go from one field to another field and bring uh the tools within so this is the how it does so this is just merging different approaches which is already common in One Direction in one field and applying to another field and then uh the Fe this is done with the prompts this is prompts and there are maybe 20 prompts which to generate uh one document but you see that um adjusting it then you can uh uh improve a lot and put more details it will put a lot so this is the short answer there it's in GitHub so get people to have editor access on the repo for the event and then have it just as a starting point and make edits in collaborative fashion and get just like anything other else okay thanks thanks a lot and rudol yeah thank you very much for this for this very interesting presentation I think that could be sort of a game changer I've I I I I will be very open I do not yet really see how that act actually works in practice but I I like i' love to try it out man I'm a little bit uh most of the questions have already be answered because I had more or less uh my questions headed into the same directions as ailles uh but I haven't got and that may be my fault I apologize because I had to go to the door briefly but how do we operationalize this during our meeting M right then we have to we have to prepare something and we have to okay and this is why I I prepared uh the coda uh link so this is a link where you already some people already joined uh the join publication because we have um publication so it is already assign topics because they are writing what they want to contribute and more important that you come already with the idea what other people are doing what is their contribution and you can already have ideas of what do you want to write in and what do you want to say about this field or maybe this and then H is the way uh to to to do it because then here it just it's like a Google Document and then you can modify it and whatever you want you just put it there text and after this we will discuss already this draft because draft is there we have also the definition of the field is also there and it is coming from me it's it's may be wrong so that's say but it can be also uh then you can modify it and Al also you can change this direction because what we will also ask a question what is the minimal uh uh version of those a AI uh all this uh intelligence in in the uh in the materials on what are the limitation of of experimental limitations so this kind of things I invite you to join C if necessary we can open more documents there and then you can contribute okay very good I do so no thanks a lot yeah that's amazing man I to me it sounds as an experiment and I apologize man see I I'm an old guy so uh learning new technologies is always hard for me but let's try it will'll be fun Vil yes uh hello um I have a quick question um these profiles that are automatically generated for all us can we modify them if we think that something is not yeah course a text file you can modify it's it will be better because if you modify your profile that will be much better because and and and and and we can and and we can access them in the website somewhere or yeah yeah yeah and it it can be done like it's in a open it can be um done with access yes okay thank you yeah just to share like a funny thing from the previous Symposium a presenter with a name that was shared with another researcher the perplexity internet search on profile was 100% incorrect so then this person looked at it sent in an updated manually written version but that's a shortterm type thing as we're just experimenting on these early iterations planning more weeks and months out and then at a deeper time scale having more updated information and Clarity around what's open and what's shared with organizers and organizations and so on about who's available for what and what's updated information on their Nano Publications and on their preferences on their capacity the more and the more relevant and accurate that information is enables this kind of decentralized knowledge collaboration using synthetic methods better so wherever accurate information can be shared it's like it'll be leveraged if the information is not in the research profile then it's not going to come into play in the later catechism based grant yeah so uh we have now few minutes before the the end and yes V just another quick one uh okay so what is the best time frame uh until we can update our profiles so the engine will have enough time to uh Pro until that's until the meeting we have roughly 8 nine days know this this you can modify this is this is just a a toy model we'll see how it works and after this we will see during the meeting maybe for sure we will have much better ideas that now okay yeah that sounds really exciting to have months in advance clear streamlined non overloading Communications have accurate information connect people and then enjoy the possibilities for having connections at events and symposia given that they're only several hours long and use the before and the during and the after to to recognize and build on all of this amazing work and people who are at different career stages there's and so many connections with how prior publishing methods existing grant opportunities new publishing methods like nanop Pub new grant opportunities yeah this will will explore a lot because European grants is not the only option but now it is becomes more and more the decentralized funding that comes from um also from this blockchain and uh decentralized science is the same so you don't need or even University to be researcher and then you can collaborate through the such tools and this is how also Daniel operate this institute is a multidisciplinary and distributed this is uh the future I think yes okay so if no more questions then uh uh we meet each other very soon almost in a week and uh we we will start with with this discussion in in in life yes okay thanks thanks a lot Vladimir thank you thank you for your time thank you very much thank you